

A

Capstone Project Report on

Pneumonia disease detection using Machine Learning:

RespiraCheck

Submitted during 4th semester in partial fulfilment of the requirements for the
award of degree of

Bachelor of Technology

in

Computer Engineering

by

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Jan-May 2026

CANDIDATE’S DECLARATION

I hereby certify that the work which is being presented in this Project titled “Pneumonia disease detection using Machine Learning: RespiraCheck” in fulfilment of the requirements for the degree of Bachelor of Technology in Computer engineering and submitted to “J.C. Bose University of Science and Technology, YMCA, Faridabad”, is an authentic record of my own work carried out under the supervision of Dr Umesh.

The work contained in this project has not been submitted to any other University or Institute for the award of any degree or diploma by me.

Student’s Signature

Vaani

CERTIFICATE

It is hereby certified that the project titled **Pneumonia disease detection using Machine Learning: RespiraCheck** submitted by **Vaani (24001003136)** of 2nd Year(4th Semester) to “**J. C. Bose University of Science and Technology, YMCA, Faridabad**” for the award of the degree of Bachelor of Technology in Computer Engineering is a record of a bonafide work carried out by her in the Project Lab – J.C. Bose University of Science and Technology, YMCA, Faridabad under my supervision as mentor for May-2026 examination.

In my opinion, the project has reached the standards of fulfilling the requirements of the regulations to the degree.

Dr. Umesh (Mentor)

Assistant Professor

Department of Computer Engg.

ACKNOWLEDGEMENT

We have taken efforts in this project. However, it would not have been possible without the kind support and help of many individuals.

I would like to extend my sincere thanks to all them. I am highly indebted to my mentor Dr. Umesh for their guidance and constant supervision as well as for providing necessary information regarding the project & also for their support in completing the project.

I would like to express my gratitude towards my parents & members of J.C. Bose University of Science and Technology for their kind cooperation and encouragement which helped in the completion of this project. I would like to express my special gratitude and thanks to my partner Tanishq as well for his support.

Student's Signature

Tanishq

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CHAPTER 1: INTRODUCTION

1.1 Introduction of the Project

Pneumonia is a serious respiratory infection that inflames the air sacs in one or both lungs, and timely diagnosis is critical for effective treatment. Chest X-rays are the most common diagnostic tool for pneumonia; however, interpreting them requires significant radiological expertise — a resource that is scarce in remote and underserved regions.

RespiraCheck addresses this gap by providing an automated, interpretable, and production-grade chest X-ray screening system with an AI-powered patient chat assistant. The system performs binary classification of chest X-ray images (NORMAL vs. PNEUMONIA) using two competing backbone architectures — DenseNet121 and EfficientNetB3 — trained with a rigorous two-phase transfer learning strategy.

1.2 Problem Statement

The traditional diagnosis of pneumonia through chest X-ray interpretation faces several critical challenges:

- **Limited Access to Expertise:** Radiologists are scarce in rural and underserved areas, leading to delayed diagnosis.
- **Inconsistency in Interpretation:** Human interpretation varies based on experience, fatigue, and workload.
- **Increasing Healthcare Burden:** Rising patient volumes demand scalable, automated screening solutions.
- **Lack of Explainability:** Black-box AI models reduce clinician trust and adoption.

Patient Communication Gap: Patients often lack accessible explanations of diagnostic results.

1.3 Objectives of the Project

- **Develop and Compare Architectures:** Implement DenseNet121 and EfficientNetB3 with identical classification heads for fair comparison.

- Implement Two-Phase Transfer Learning:
 - Phase 1: Frozen backbone, train classification head (LR=1e-4)
 - Phase 2: Unfreeze last 30 layers, fine-tune (LR=1e-5)
- Automate Model Selection: Select best model based on AUC-ROC metric on held-out test set.
- Integrate Grad-CAM++ Explainability: Generate visual heatmaps highlighting regions influencing predictions.
- Deploy Full-Stack Application: Flask REST API (/predict, /chat, /health) + React frontend.
- Integrate AI Chat Assistant: Gemini 2.5 Flash for patient education and result explanation.
- Achieve Performance Targets: Test accuracy >89%, Pneumonia recall >93%, AUC-ROC >0.95. Replace Binary Cross-Entropy with Binary Focal Cross-Entropy ($\alpha=0.25$, $\gamma=2.0$) for improved handling of class imbalance.

1.4 Learning from the Project

- Advanced Transfer Learning Strategies (Freezing/Unfreezing, Layer-wise LR)
- Model Comparison & Automated Selection Pipelines
- Explainable AI (Grad-CAM++) Implementation
- Full-Stack Development (Flask + React + CORS + Environment Management)
- AI Chatbot Integration with Context-Aware Prompt Engineering
- Medical AI Ethics & Safety Guidelines Implementation

CHAPTER 2: LITERATURE REVIEW / RELATED WORK

2.1 Related Studies or Existing Systems

The application of machine learning and deep learning to medical image analysis, particularly pneumonia detection, has been extensively researched:

CNN Architectures in Medical Imaging:

- VGG Networks: Early work demonstrated CNN feasibility but suffered from high parameter counts.
- ResNet Architecture: Introduced skip connections enabling deeper networks; used in Semester 3 baseline.
- DenseNet Models: Feature reuse via dense connections improves gradient flow and parameter efficiency.
- EfficientNet: Compound scaling optimizes depth/width/resolution for optimal accuracy-efficiency tradeoff.

Transfer Learning in Medical Imaging:

Studies consistently show transfer learning from ImageNet pre-trained weights improves accuracy by 5-15% compared to training from scratch on small medical datasets (Rajpurkar et al., 2017; Wang et al., 2020).

Explainability Methods:

- Grad-CAM (Selvaraju et al., 2017): Class activation mapping via gradients.
- Grad-CAM++ (Chattopadhyay et al., 2018): Higher-order gradients improve localization for multiple object instances.

Existing Systems:

- qXR by Qure.ai: Commercial FDA-cleared chest X-ray interpretation system.
- ChestX-ray8 (NIH): Large-scale dataset enabling benchmark research.
- CheXNet (Stanford): 121-layer DenseNet achieving radiologist-level pneumonia detection.

Recent Advances:

Xu & Zhang (Scientific Reports, 2026) proposed a Double SGAN augmentation framework with spectral normalization and self-attention to address class imbalance in pneumonia datasets — an approach under investigation for future RespiraCheck iterations.

CHAPTER 3: METHODOLOGY

3.1 System/Model Architecture

The overall pipeline ingests a chest X-ray image, preprocesses it through resizing and normalization, passes it through one of the two backbone architectures (DenseNet121 or EfficientNetB3), and produces a binary NORMAL/PNEUMONIA classification with associated confidence score and Grad-CAM++ heatmap.

Model Architecture (Shared Classification Head):

Both architectures share an identical classification head: GlobalAveragePooling2D → Dense(512, ReLU) → Dropout(0.5) → Dense(1, Sigmoid). This ensures any performance difference is attributable solely to backbone architecture. Both models are compiled with BinaryFocalCrossentropy($\alpha=0.25$, $\gamma=2.0$, apply_class_balancing=True) as the loss function, replacing the Binary Cross-Entropy used in the previous semester's prototype. Two Phase training strategy is used, in first state the backbone state is fully frozen and learning state is set to $1e-4$ and 10 epochs are run

to get trained classification head. In next step fine tuning is done by unfreezing the last 30 layers and using the learning rate as $1e-5$. Both models are evaluated on the test set; the winner is selected by AUC-ROC and the path is written to best_model_path.txt for API startup.

3.2 Dataset / Data Collection

Source: Kaggle "Chest X-Ray Images (Pneumonia)" dataset (Guangzhou Women and Children's Medical Center).

Dataset Distribution:

Split	Normal	Pneumonia	Total
Training	1,341	3,875	5,216
Validation	8	8	16
Testing	234	390	624
Total	1,583	4,273	5,856

Preprocessing Pipeline:

- Re-splitting: Merged train/val, performed stratified 80/20 split (random_state=42)
- Resizing: 224×224 pixels for both architectures

- Normalization: Pixel values scaled to [0,1]
- Augmentation (training only): Random rotation $\pm 20^\circ$, horizontal flip, zoom 10%, width/height shift 10%

3.3 Tools and Technologies Used

Category	Technology / Library
Programming Language	Python 3.10+
Deep Learning Framework	TensorFlow 2.20.0, tf_keras 2.20.0
Pre-trained Architectures	DenseNet121, EfficientNetB3 (ImageNet weights)
Explainability	Grad-CAM++ via tf-keras-vis 0.8.7
AI Chat Assistant	Google Gemini 2.5 Flash (google-genai 1.68.0)
Data Handling	NumPy 2.2.4, Pandas 2.2.3, Scikit-learn 1.6.1
Image Processing	OpenCV 4.11.0, Pillow 11.2.1
Visualization	Matplotlib 3.10.1, Seaborn 0.13.2
Backend	Flask 3.1.0, Flask-CORS 5.0.1, python-dotenv 1.2.2
Frontend	React 18.3.1, Vite 6.3.5, JavaScript/TypeScript
Dataset	Kaggle: Chest X-Ray Images (Pneumonia) — 5,856 images
Dev Environment	VS Code, Google Colab, Jupyter Notebook

3.4 Algorithms / Techniques Implemented

- Transfer Learning: Leveraging ImageNet pre-trained weights for feature extraction
- Two-Phase Training: Frozen backbone → selective fine-tuning
- Class Imbalance Handling: Binary Focal Cross-Entropy Loss ($\alpha=0.25$, $\gamma=2.0$) with `apply_class_balancing=True` - automatically down-weights easy examples and focuses training on hard, misclassified samples, replacing manual sklearn class weight computation.
- Data Augmentation: Geometric and photometric transforms to improve generalization
- Early Stopping & Learning Rate Scheduling: Prevent overfitting, optimize convergence
- Grad-CAM++: Higher-order gradient-based visualization for model interpretability
- Context-Aware Prompt Engineering: Safe, empathetic Gemini chatbot responses
- Automated Evaluation Pipeline: Metrics computation, model comparison, selection

CHAPTER 4: IMPLEMENTATION

4.1 System Design

The system follows a modular architecture comprising seven independent modules: Data Preprocessing, Model Building, Training, Evaluation, Explainability, Backend API, and Frontend. Each module exposes well-defined interfaces, enabling isolated development, testing, and replacement.

4.2 Module Description

Module 1: Data Preprocessing (`data_setup.py`)

- Merges train/val directories, performs stratified 80/20 re-split
- Computes balanced class weights using `sklearn.utils.class_weight`
- Saves file lists and weights as pickle files for reproducibility

Module 2: Model Building (`model.py`)

- `build_phase1_model()`: DenseNet121 with frozen backbone
- `build_phase2_model()`: Unfreezes last 30 layers for fine-tuning
- `build_efficientnet_phase1/2_model()`: Identical pipeline for EfficientNetB3
- Shared classification head (GAP $\rightarrow$ Dense(512, ReLU) $\rightarrow$ Dropout(0.5) $\rightarrow$ Sigmoid) ensures architecture-only comparison.

Module 3: Training (`train.py`)

- Sequential training: DenseNet Phase1 $\rightarrow$ 2, then EfficientNet Phase1 $\rightarrow$ 2
- Callbacks: `EarlyStopping(monitor=val_loss, patience=5)`, `ModelCheckpoint(monitor=val_auc, save_best_only=True)`, `ReduceLROnPlateau(factor=0.5, patience=3, min_lr=1e-7)`.
- Saves training history, model checkpoints, and comparison plots

Module 4: Evaluation (`evaluate.py`)

- Computes: Accuracy, Precision, Recall, F1, AUC-ROC, Average Precision
- Generates: Confusion matrices, ROC curves, PR curves
- Writes best model path to `best_model_path.txt` based on AUC-ROC

Module 5: Explainability (`gradcam.py`)

- generate_gradcam(): Native TensorFlow implementation using nested GradientTapes
- overlay_heatmap(): JET colormap blending with original image
- Handles fallback to standard Grad-CAM if higher-order gradients unavailable

Module 6: Backend API (app.py)

Endpoints:

- POST /predict: Image upload → inference → Grad-CAM → JSON response
- POST /chat: Gemini-powered medical assistant with prediction context
- GET /health: Liveness probe for deployment monitoring

Features: File validation, error handling, environment-based config, model auto-loading

Module 7: Frontend (frontend/)

Components:

- Drag-and-drop image upload with preview
- Real-time prediction display with confidence score
- Grad-CAM++ heatmap overlay visualization
- Integrated chat window with markdown rendering

Styling: Glassmorphism UI, responsive design, accessibility considerations

CHAPTER 5: RESULTS AND DISCUSSION

5.2 Evaluation Metrics

DenseNet121 (Selected Model) Performance:

Metric	Normal	Pneumonia	Overall
Precision	0.89	0.94	0.92
Recall	0.85	0.9231	0.89
F1-Score	0.87	0.9314	0.90
Support	234	390	624

Model Comparison:

Metric	DenseNet121	EfficientNetB3	Winner
Test Accuracy	89.58%	87.21%	DenseNet121
Pneumonia Recall	93.59%	90.15%	DenseNet121

Metric	DenseNet121	EfficientNetB3	Winner
AUC-ROC	0.9568	0.9412	DenseNet121
Avg Precision	0.9707	0.9634	DenseNet121
F1-Score (Pneumonia)	0.9314	0.9187	DenseNet121

Confusion Matrix – DenseNet121:

	Predicted Normal	Predicted Pneumonia
Actual Normal	194 (TN)	40(FP)
Actual Pneumonia	25 (FN)	365 (TP)

5.3 Results Analysis

Strengths:

- High Pneumonia Recall (93.59%): Excellent sensitivity — critical for medical screening to minimize false negatives. Strong AUC-ROC (0.9549): Indicates robust discrimination ability across classification thresholds.
- Explainability Integration: Grad-CAM++ heatmaps provide visual justification for predictions, increasing clinician trust.
- Patient-Centric Design: Gemini chat assistant bridges the gap between technical output and patient understanding.

Clinical Relevance:

- The system is designed as a pre-diagnostic screening tool, not a replacement for radiologists.
- High recall prioritizes catching potential pneumonia cases for expert review.
- Explainability features support clinician-AI collaboration rather than black-box automation.

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 Conclusion

RespiraCheck successfully develops an automated pneumonia detection system that significantly advances beyond the Semester 3 ResNet50 prototype. Key achievements include:

- **Superior Model Performance:** DenseNet121 achieved 89.58% test accuracy, 93.59% pneumonia recall, 0.9568 AUC-ROC, and 0.0439 test loss — exceeding all stated targets. The adoption of Binary Focal Cross-Entropy loss ($\alpha=0.25$, $\gamma=2.0$) improved overall model calibration and replaced manual class weight computation with a principled, loss-level approach to class imbalance.
- **Rigorous Methodology:** Two-phase transfer learning with automatic model selection ensures reproducible, optimal architecture choice.
- **Explainability by Design:** Grad-CAM++ integration provides visual interpretability critical for medical AI adoption.
- **Full-Stack Deployment:** Flask REST API + React frontend delivers a usable, accessible application for non-technical users.
- **Patient Education:** Gemini 2.5 Flash chat assistant contextualizes results in empathetic, accessible language while maintaining safety boundaries.

6.2 Limitations

- **Dataset Constraints:** Performance validated only on Kaggle dataset; real-world generalization requires multi-center validation.
- **Binary Classification:** Cannot distinguish pneumonia subtypes (bacterial/viral) or other thoracic conditions.
- **Explainability Scope:** Grad-CAM++ highlights regions but does not provide pathological reasoning.

6.3 Future Scope

Model Improvements:

- Extend training to 20-30 epochs with layer-wise learning rate decay
- Implement hyperparameter optimization (Optuna, Ray Tune)

Enhanced Functionality:

- **Multi-class Classification:** Extend to bacterial/viral pneumonia, COVID-19, TB using NIH ChestX-ray14 dataset

- Pneumonia Localization: Train YOLOv8 detection model on RSNA bounding box annotations

Application Development:

- Professional UI redesign with React component library (Material-UI, Chakra UI)
- Mobile application using React Native + TensorFlow Lite for offline inference

Clinical Validation:

- Conduct reader studies comparing AI-assisted vs. unassisted radiologist performance
 - Pursue regulatory pathways (CE marking, FDA SaMD classification)
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